

D1 Decision Brief

D1 — Archetype-Routed Kill Rule · Decision Brief

Date: 2026-06-03 · **For:** Hume **Status:** Awaiting one decision before build · **Stakes:** changes a live trade gate (conviction + position size)

The decision in one line: D1 loosens the "this trade is broken" gate for regime-shift names only. You are choosing *how* to enforce that — **who applies the threshold** (code vs the prompt) and **how far it reaches** (the thesis path now, or also the live Socratic path). The thresholds themselves are already settled.

1. What the kill rule does today

Every thesis computes `risk_adj_ev_ratio = risk_adj_target / current_price` — the probability-weighted exit price divided by today's price. A fixed table in `prompts/thesis_v3.md` then clamps the trade:

`risk_adj_ev_ratio < 0.90` → `conviction: BROKEN, position_size_pct: 0%` — regardless of how strong the strategic thesis is.

In plain terms: *if the math says your probability-weighted exit is more than ~10% below where you'd buy, the trade is dead*. This rule exists for a real and good reason — the **AXON / COHR / ALAB pattern**: the secular story is correct, but the multiple is already at peak, so the asymmetry is inverted and you shouldn't buy. The gate catches exactly that.

2. The problem it now causes

The same uniform 0.90 gate mis-fires on **regime-shift / transformational names**. Those names are valued deliberately conservatively — and as of V2 (shipped this session) they use **DCF-as-a-floor**, which by design produces a cautious `risk_adj_target`. So a genuinely strong regime-shift bet can show a ratio below 0.90 and get stamped `BROKEN / 0%` — not because the trade is bad, but because the math under it is intentionally cautious.

LITE is the live example: a high-conviction AI-photonics regime-shift name that the uniform gate would mark broken on conservative EV math alone. That's a false kill, and it's the motivating case in the backlog (D1's pass criterion is "re-run on LITE and confirm it no longer fires BROKEN").

3. The fix

Route the kill threshold by archetype instead of applying one number to everyone:

Archetype group	Kill threshold	Rationale
cyclical / garp / compounder / special-situation	0.90 (unchanged)	The gate works as designed for these.
transformational (regime-shift)	0.60 (wider)	Only kill when the math is <i>really</i> bad, not merely conservative.
pre-revenue ($\approx$ zero TTM revenue)	gate disabled	The ratio is meaningless with no revenue/EV anchor.

These thresholds come straight from the backlog and are **not** part of the open decision — they're settled.

4. The actual decision — two dials

Dial 1 — who enforces the threshold?

- **In code (Python post-processor) — recommended.** After the model returns, code re-routes: if a trade was forced `BROKEN` purely by the ratio gate but the archetype's wider threshold isn't breached, relax it; for pre-revenue, skip the gate. Deterministic, unit-testable, and **ungameable**. The changelog already committed to exactly this ("V3.4.3.1 Python post-processor").
- **In the prompt.** Inject the archetype-specific threshold table into the prompt so the model reasons with the right gate. The prose stays coherent, but this is the *same gate already flagged as gameable* (a small probability nudge clears it), and the outcome can't be unit-tested.

Dial 2 — how far does it reach?

- **Thesis path only (`run_thesis`) — recommended for now.** Low blast radius; it's the older single-model path and doesn't disturb the live opus-4-8 judgment layer we just upgraded.
- **Also the Socratic path.** Apply the same routing to the live Model-B regime judgment. Broader and higher impact (Socratic is the active path), but it must reconcile with Model B's existing regime handling to avoid double-counting — so it belongs in its own deliberate step, not bundled here.

5. The full option space (for completeness)

The two dials above collapse five concrete implementations:

1. **Python post-processor, thesis only** — *recommended*. Testable, deterministic, low risk.
2. **Parameterize the prompt** — coherent prose, but gameable and untestable.
3. **Post-processor + Socratic** — consistent across both paths, but touches the live layer; higher stakes.
4. **Hybrid (prompt states the rule, code enforces it)** — coherent + guaranteed, but most work and two places to maintain.

5. **Move the clamp fully into Python** (model emits raw inputs; one `apply_kill_gate` is the sole source of truth) — cleanest long-term and removes the gameable prompt gate entirely, but the biggest change and it alters the contract.

6. Recommendation

Build #1 now, in the spirit of #5. A pure, exhaustively-tested `apply_kill_gate(parsed, archetype)` that is the deterministic source of truth for the ratio clamp, wired into `run_thesis`, thesis-path only. Reasoning:

- The project already decided the gate should be enforced in Python and is ungameable — that rules out the prompt-only route.
- A kill clamp is a **Hard Gate**; by your own filter philosophy it should be a deterministic rule, not model discretion.
- Scoping to the thesis path keeps the blast radius small and leaves the just-upgraded Socratic layer untouched. The Socratic version (#3) is the right *next* step, done deliberately.

7. The one subtlety to get right

Coherence. When the post-processor relaxes a `BROKEN`, the model's own prose still argued `BROKEN` under the old 0.90 table — so the override must be annotated in `kill_triggers`, e.g. *"uniform gate said BROKEN at 0.90; routed to LOW for transformational — ratio $0.78 \geq 0.60$ archetype floor."* Otherwise the sealed record contradicts itself. This is handled in the post-processor, not left implicit.

8. Validation

The pure `apply_kill_gate` logic gets full unit-test coverage here (every archetype × ratio band, plus the pre-revenue case). The behavioural confirmation is yours: one live Opus thesis run on **LITE**, expected to no longer fire `BROKEN` / 0% — the backlog's stated pass criterion.

9. After you decide

If you confirm #1: I build and test `apply_kill_gate`, wire it into `run_thesis` with the annotation, add the unit suite, and write the CHANGELOG entry — then hand you the one-line live-validation command. If you'd rather a different dial setting, the same plan adapts. If you'd rather hold D1, the next move on the track is the Opus cost cap.

Source: `prompts/thesis_v3.md` (kill table, lines ~166–181), `scripts/run_thesis.py` (post-parse hook), `scripts/target_engine.py` (archetype + V2 dcf_role routing), and the D1 entry in `data/todo_master_2026_05_26.md`.